

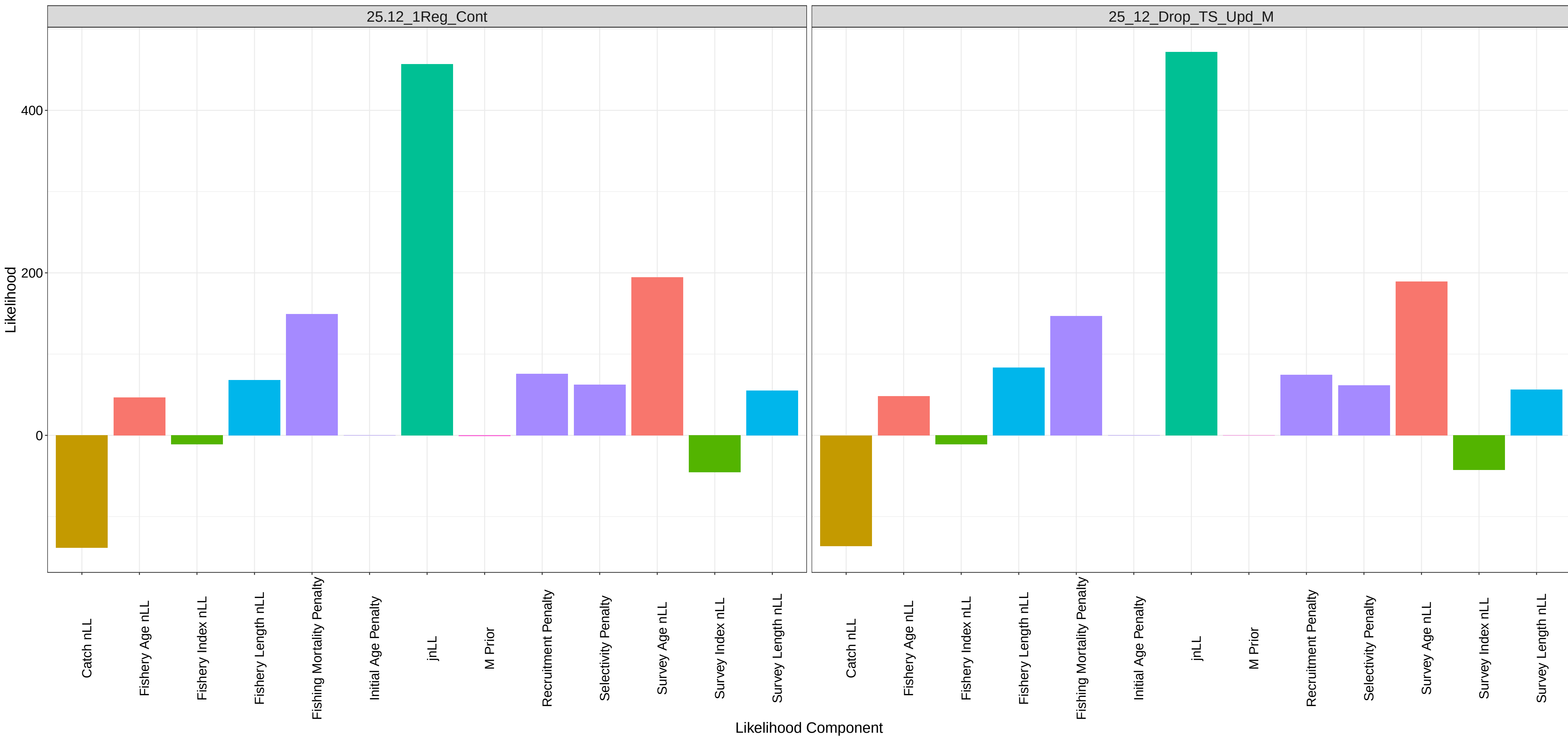
|   | Model               | Region | Terminal_SSB | Terminal_SSB0 | Terminal_F | Catch_Advice | B_Ref_Pt | F_Ref_Pt | B_over_B_Ref | B_over_DynB_Ref | F_over_F_Ref |
|---|---------------------|--------|--------------|---------------|------------|--------------|----------|----------|--------------|-----------------|--------------|
| 1 | 25.12_1Reg_Cont     | 1      | 206.6999     | 357.0208      | 0.04784668 | 39.55138     | 131.4549 | 0.07875  | 1.57240      | 0.57896         | 0.60758      |
| 2 | 25_12_Drop_TS_Upd_M | 1      | 194.0907     | 324.8450      | 0.04842486 | 39.55329     | 126.6058 | 0.08308  | 1.53303      | 0.59749         | 0.58289      |

|   | Model               | jnLL     | M Prior    | Recruitment Penalty | Selectivity Penalty | Catch nLL | Fishing Mortality Penalty | Survey Index nLL | Fishery Index nLL | Initial Age Penalty | Survey Age nLL | Fishery Age nLL | Survey Length nLL | Fishery Length nLL |
|---|---------------------|----------|------------|---------------------|---------------------|-----------|---------------------------|------------------|-------------------|---------------------|----------------|-----------------|-------------------|--------------------|
| 1 | 25.12_1Reg_Cont     | 456.8247 | -1.0189779 | 75.86092            | 62.56423            | -138.3008 | 149.3833                  | -45.39642        | -11.21223         | 0.2741517           | 194.6365       | 46.66608        | 55.19863          | 68.16928           |
| 2 | 25_12_Drop_TS_Upd_M | 471.8339 | -0.2370536 | 74.67764            | 61.83381            | -136.2188 | 146.9955                  | -42.64644        | -10.95465         | 0.2847093           | 189.5774       | 48.50805        | 56.40044          | 83.61324           |

# Negative Log-Likelihood

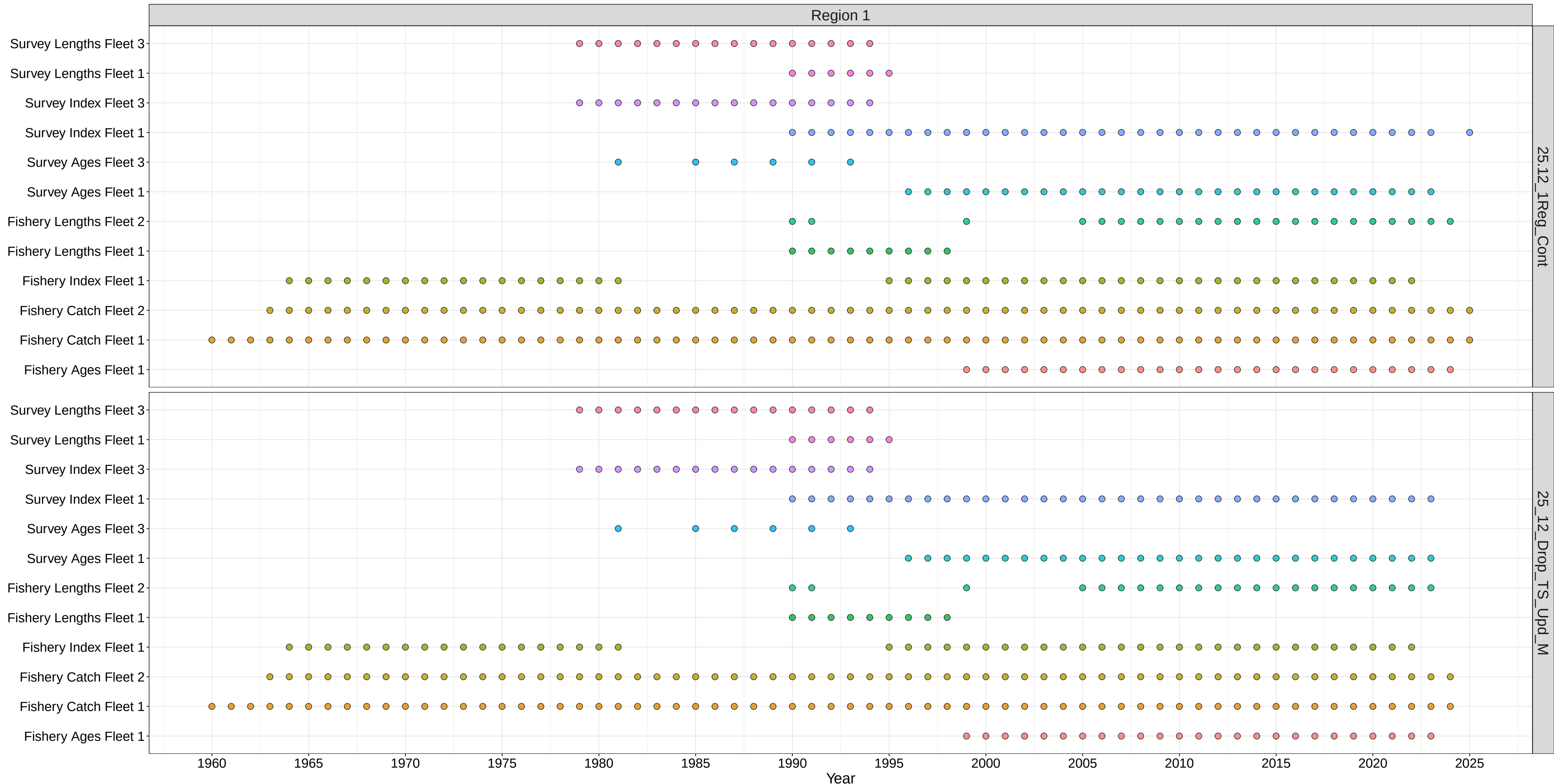
Type

- Age
- Catch
- Index
- jnLL
- Length
- Penalty
- Prior



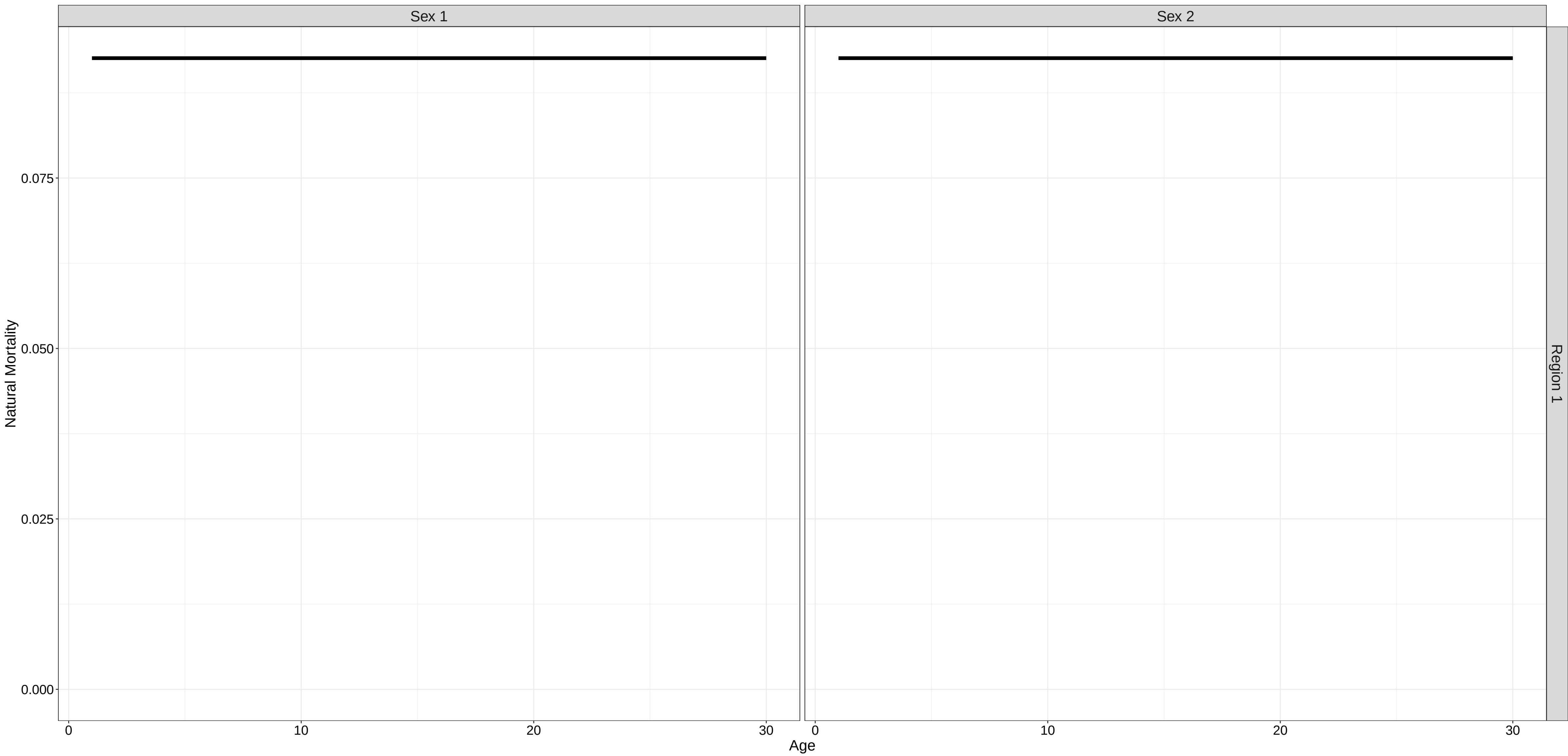


## Data



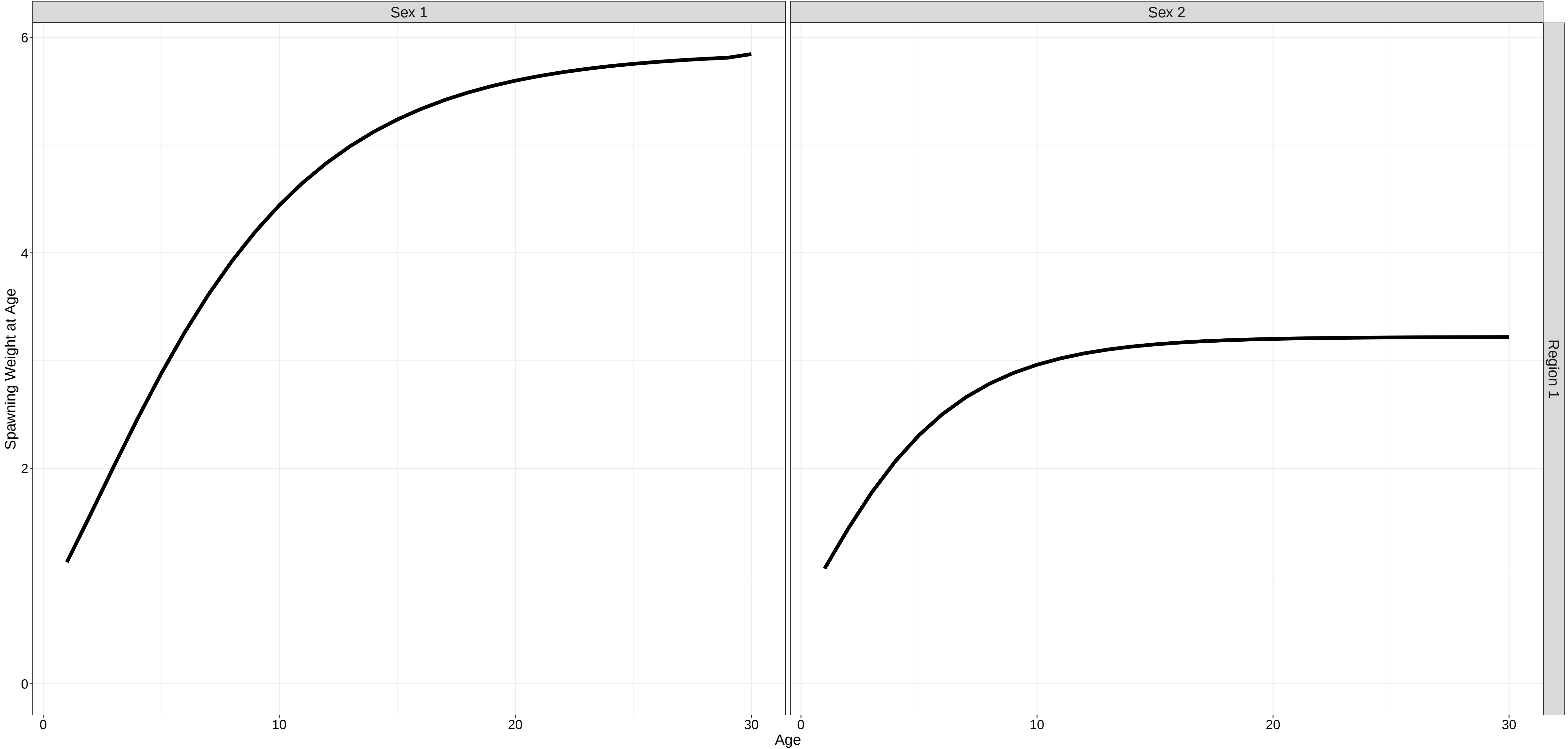
# Natural Mortality

Model 25.12\_1Reg\_Cont



Weight-at-Age

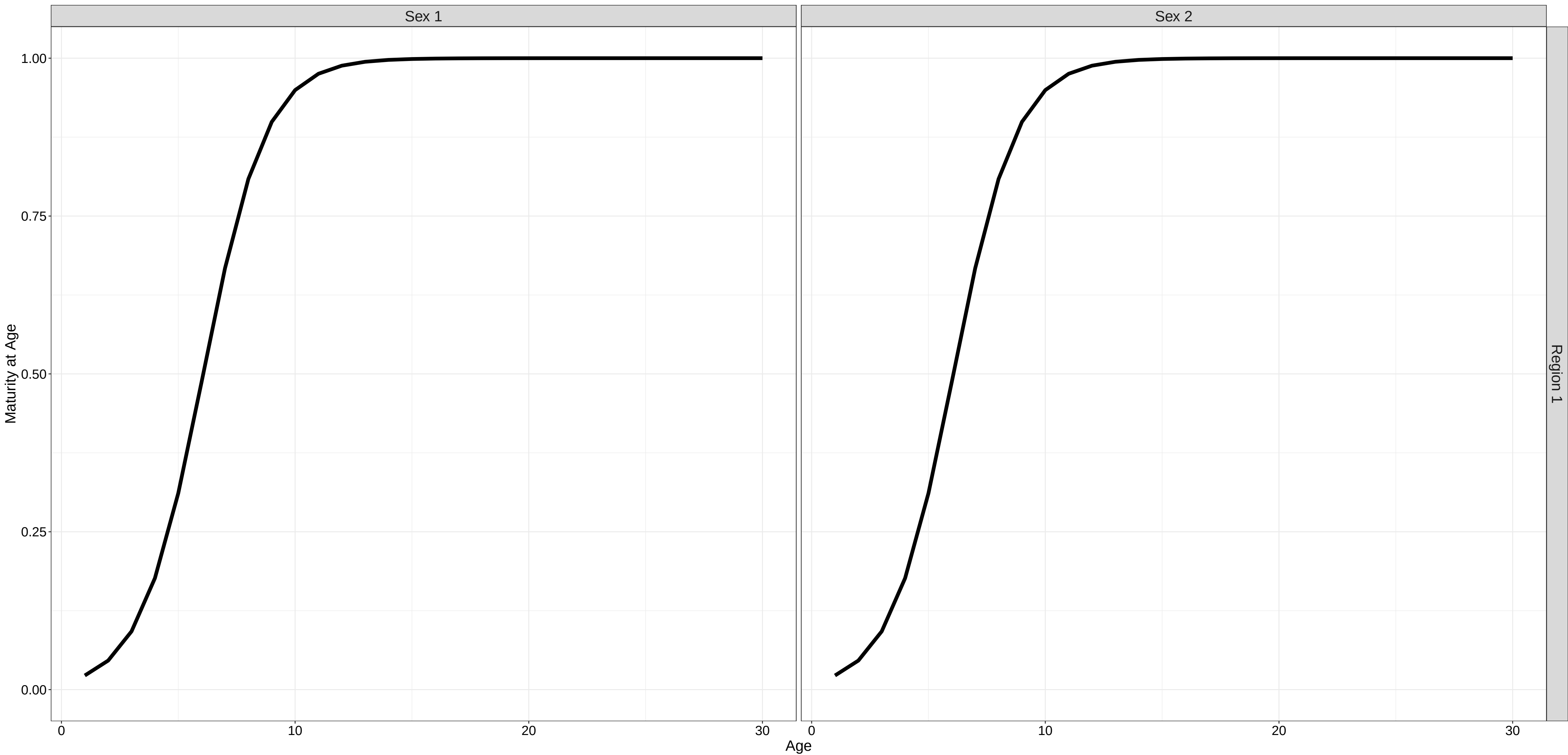
Model 25.12\_1Reg\_Cont





Maturity

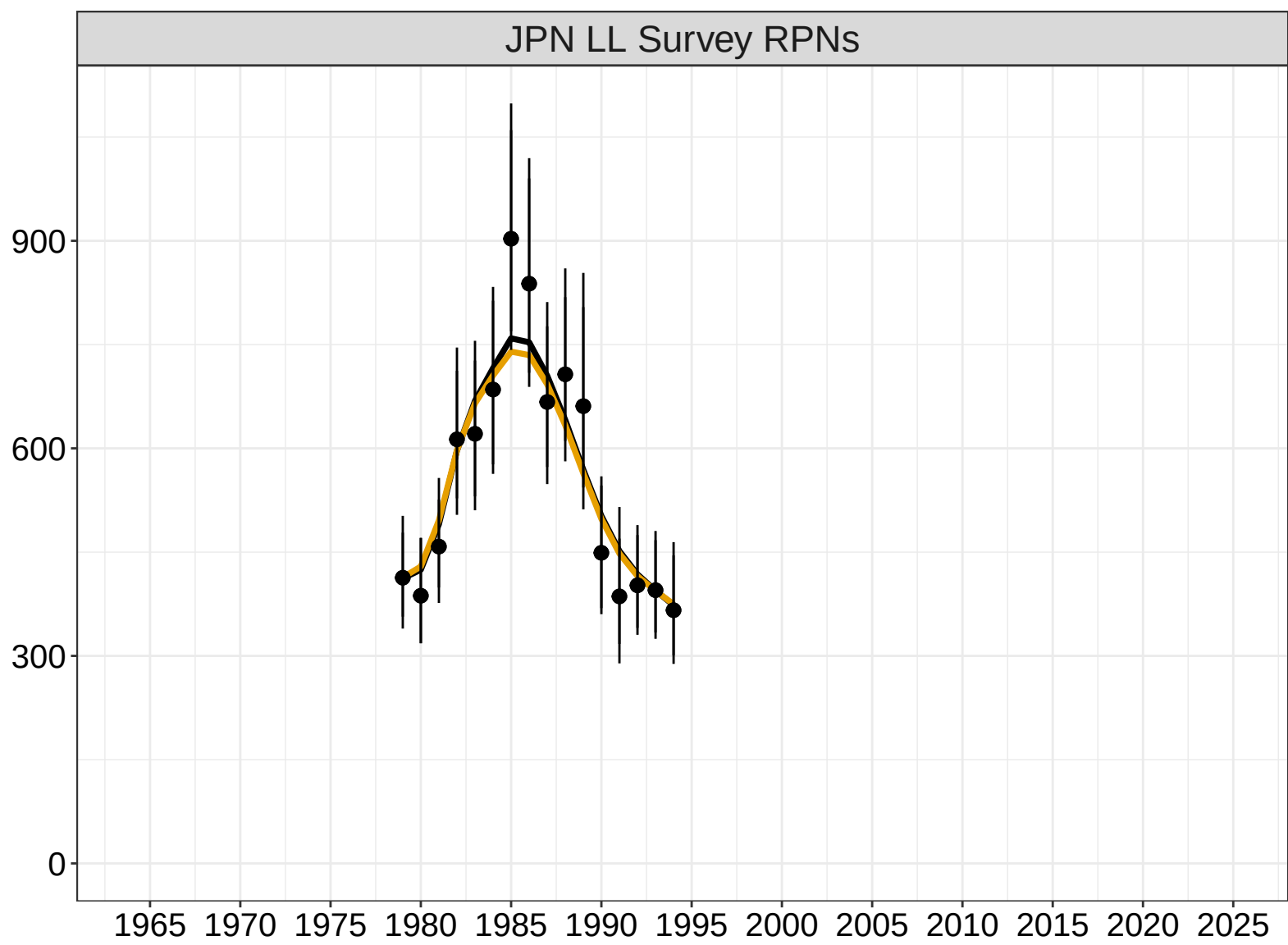
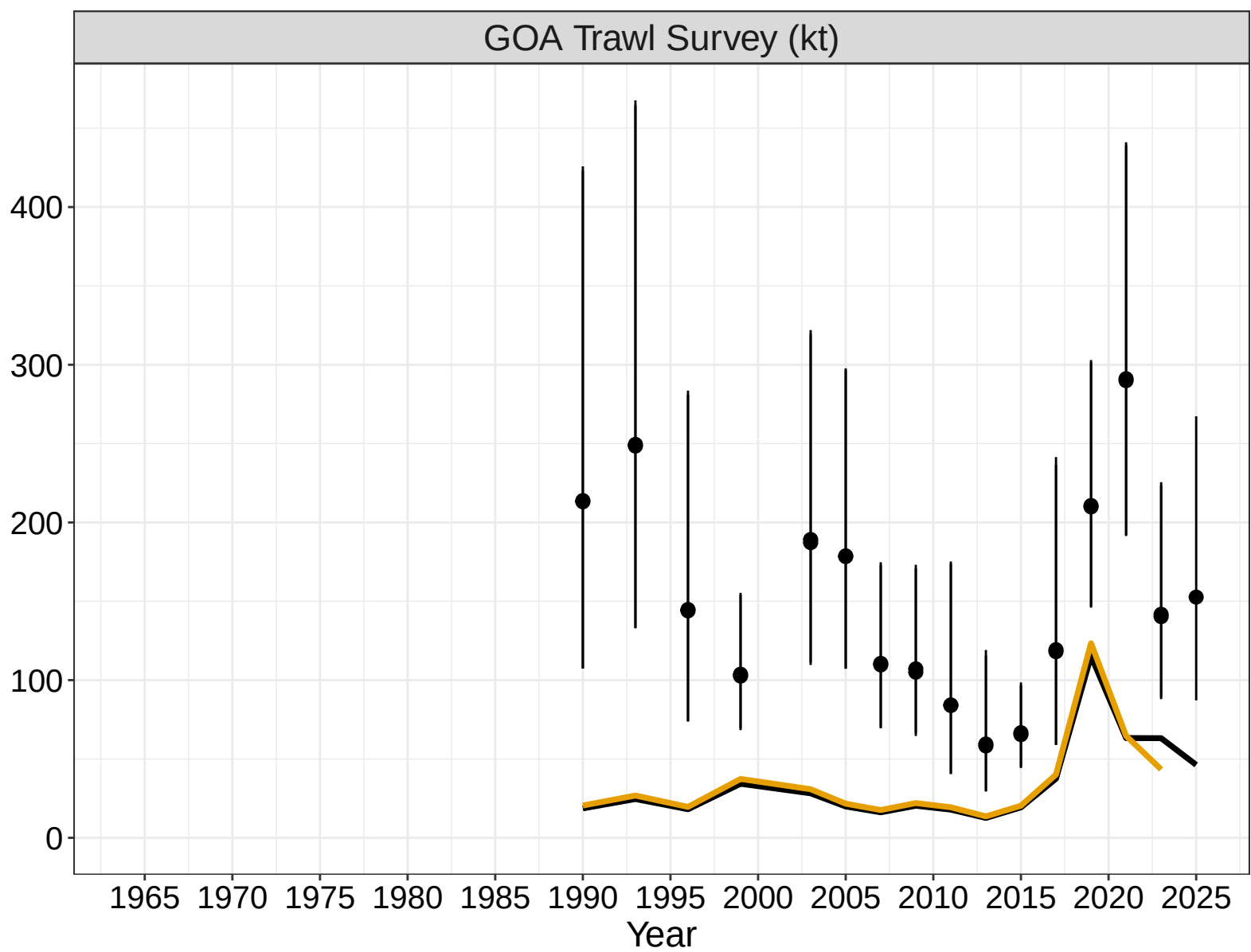
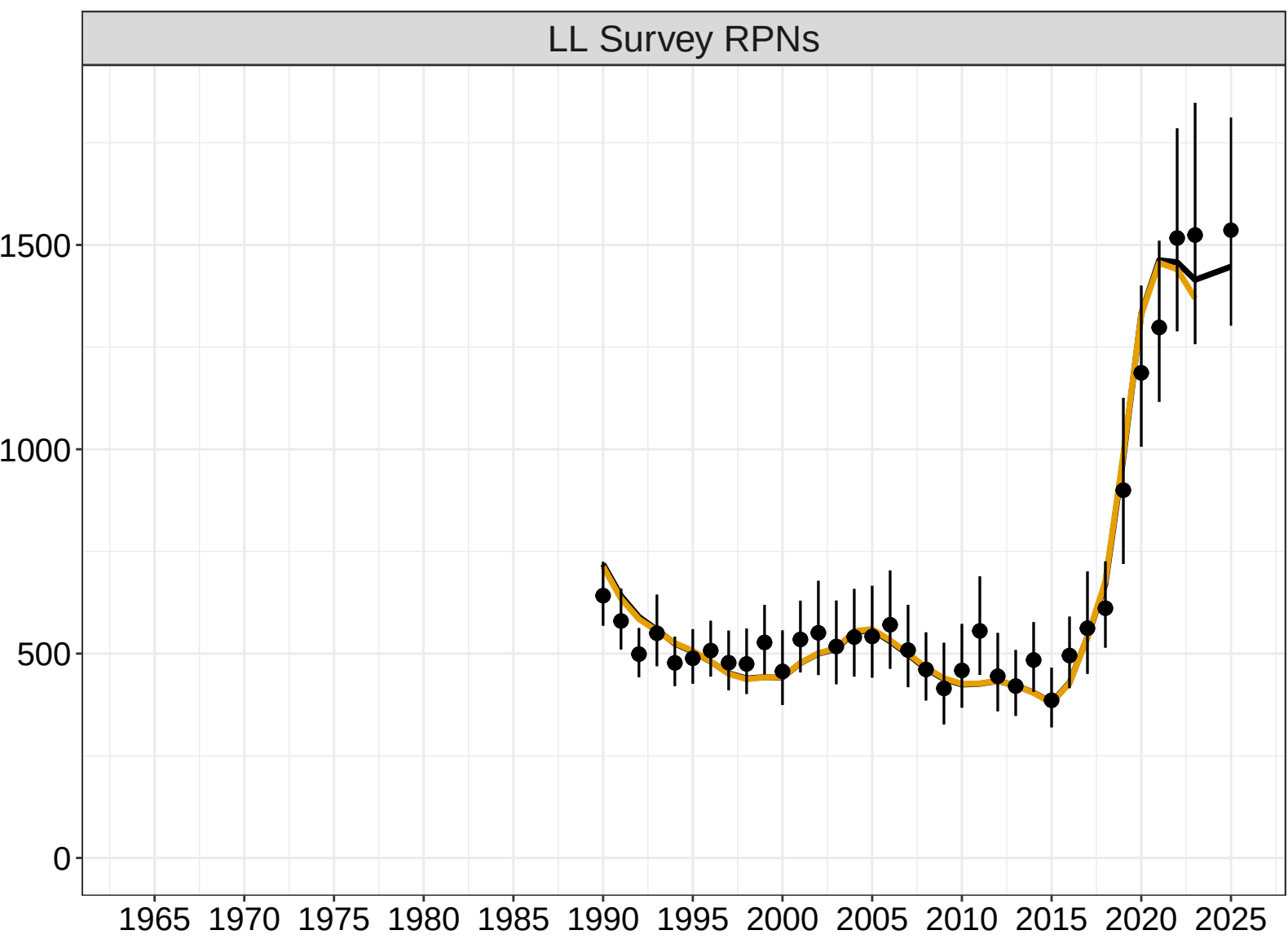
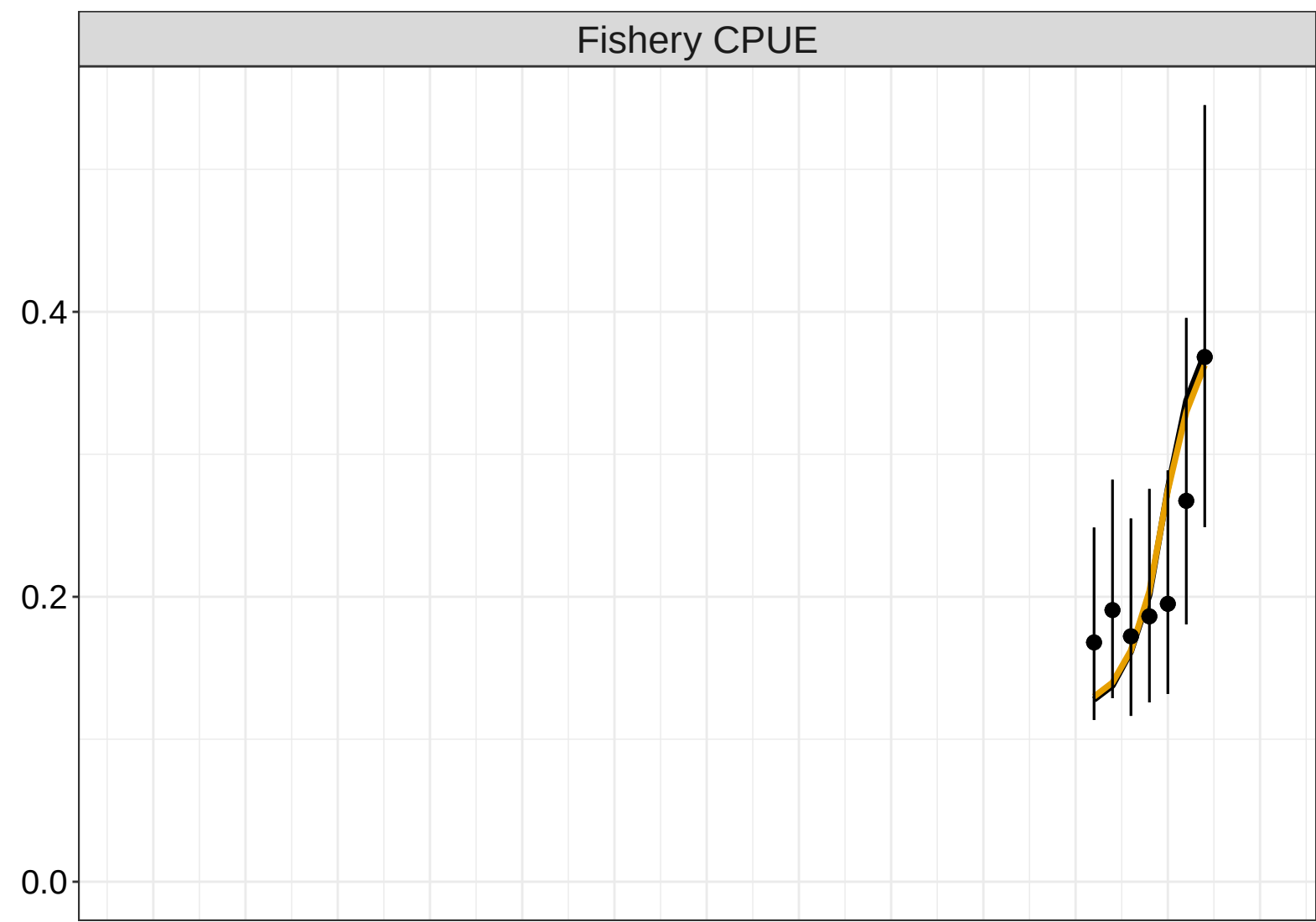
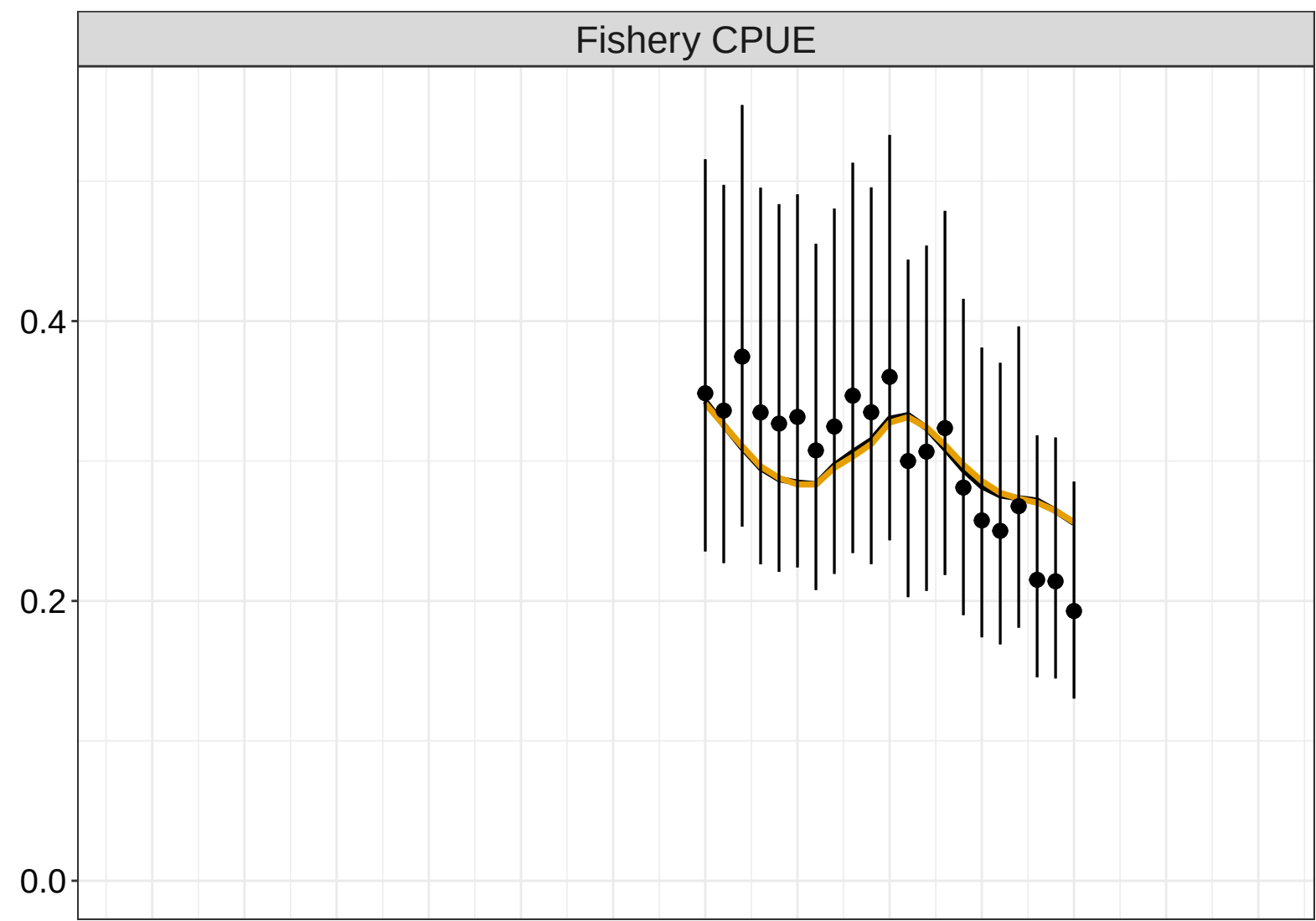
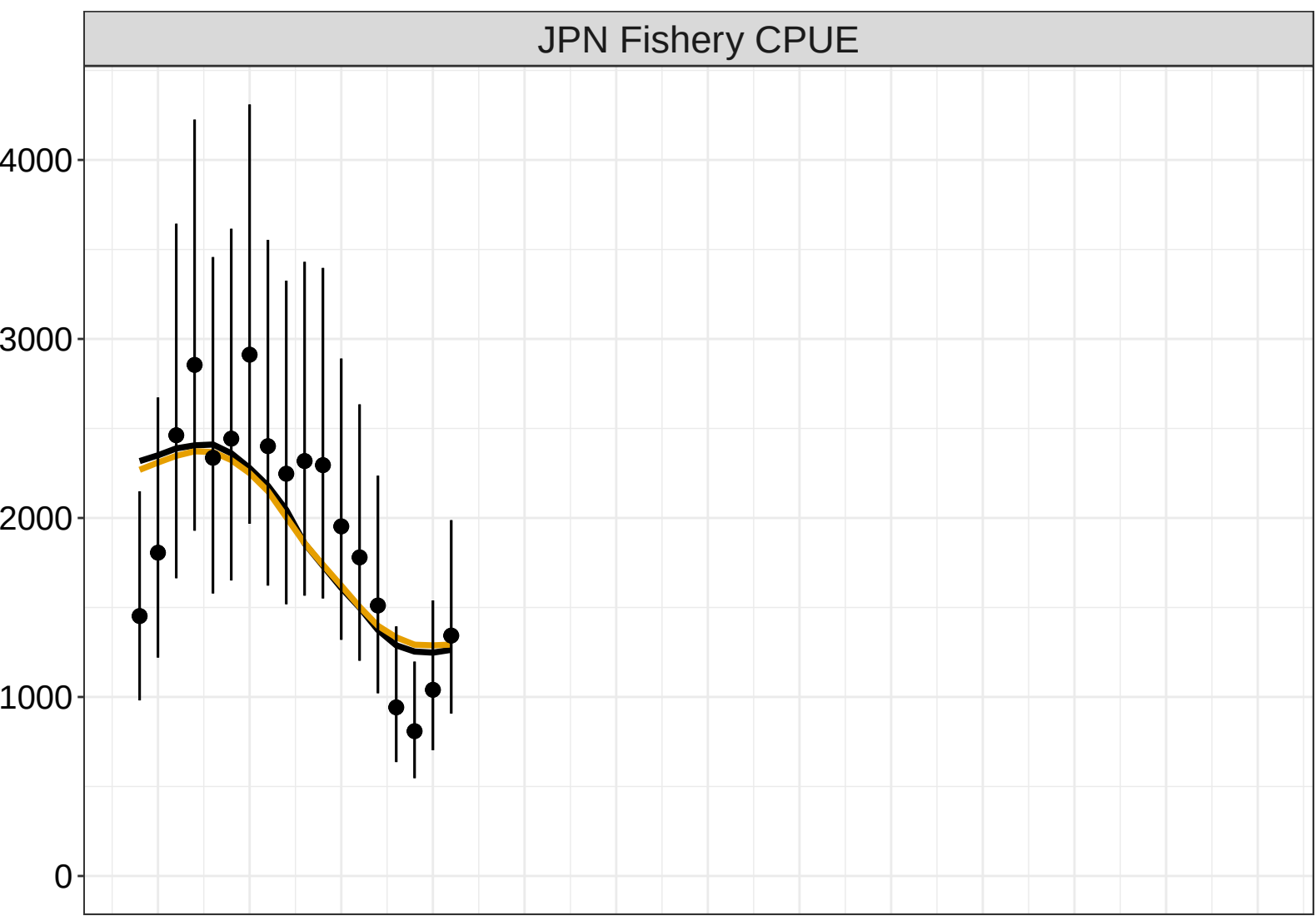
Model 25.12\_1Reg\_Cont



# Index Fits

Model — 25.12\_1Reg\_Cont — 25.12\_Drop\_TS\_Upd\_M

Index

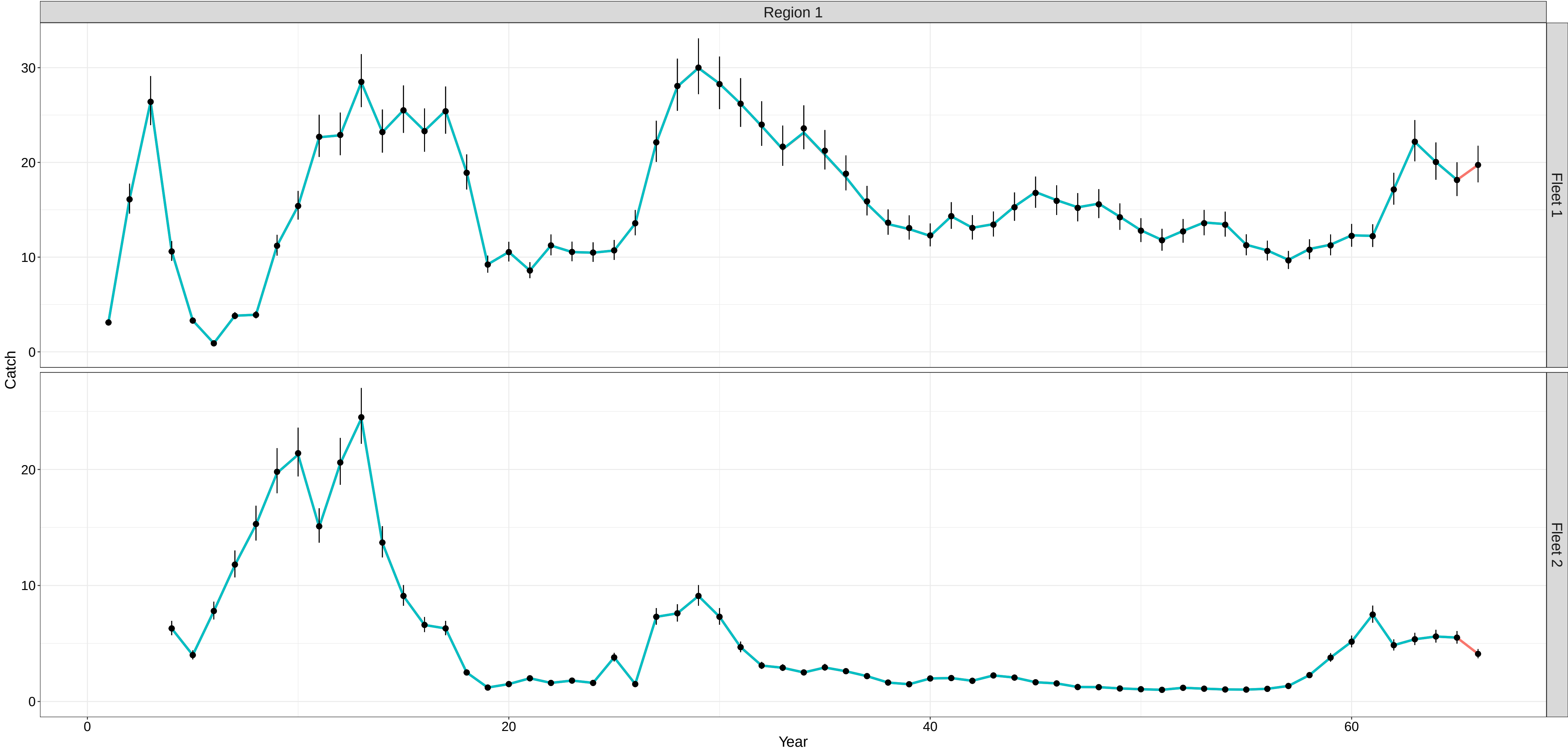


Year



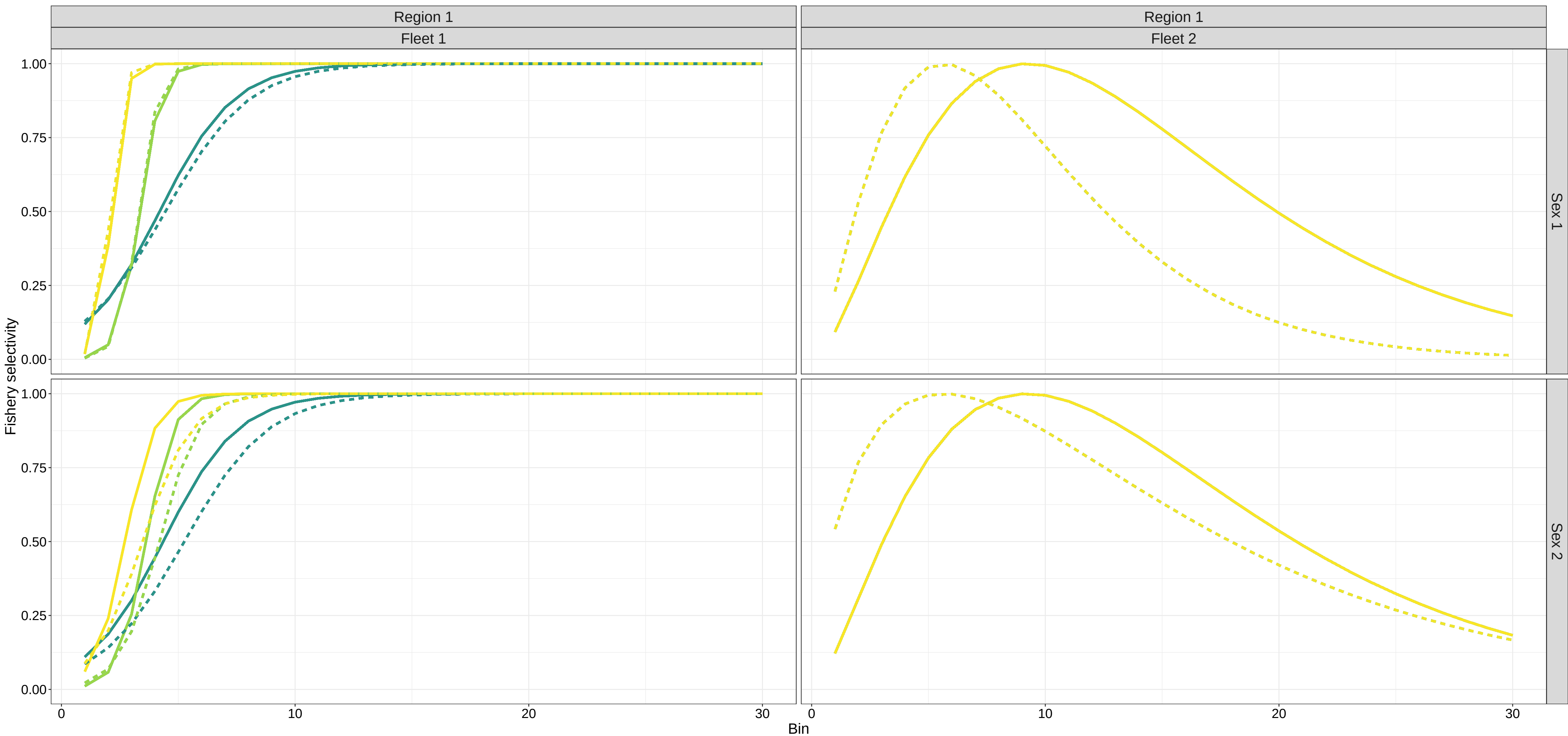
Catch

Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M



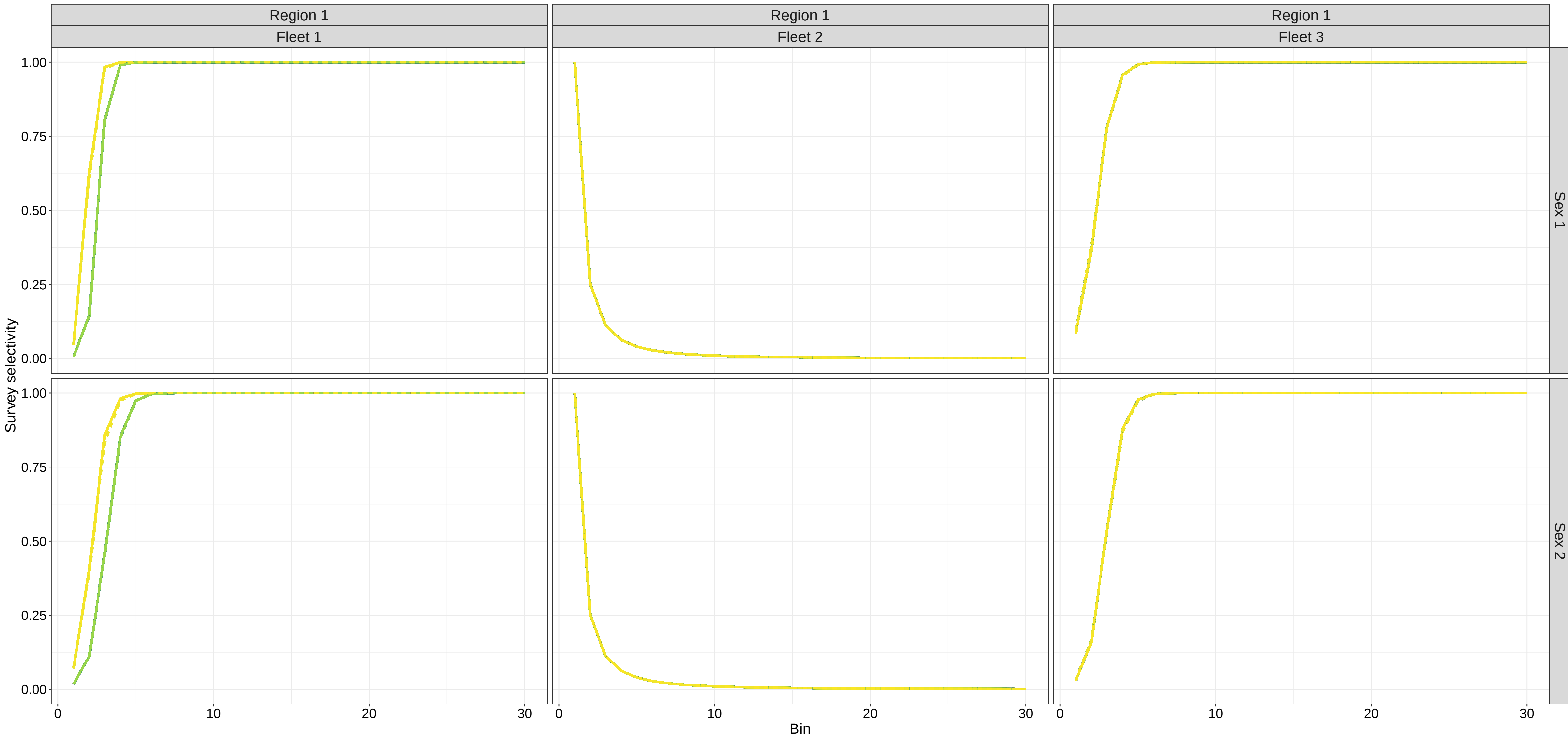
# Fishery Selectivity

Model — 25.12\_1Reg\_Cont - - - 25\_12\_Drop\_TS\_Upd\_M Year 20 40 60



# Survey Selectivity

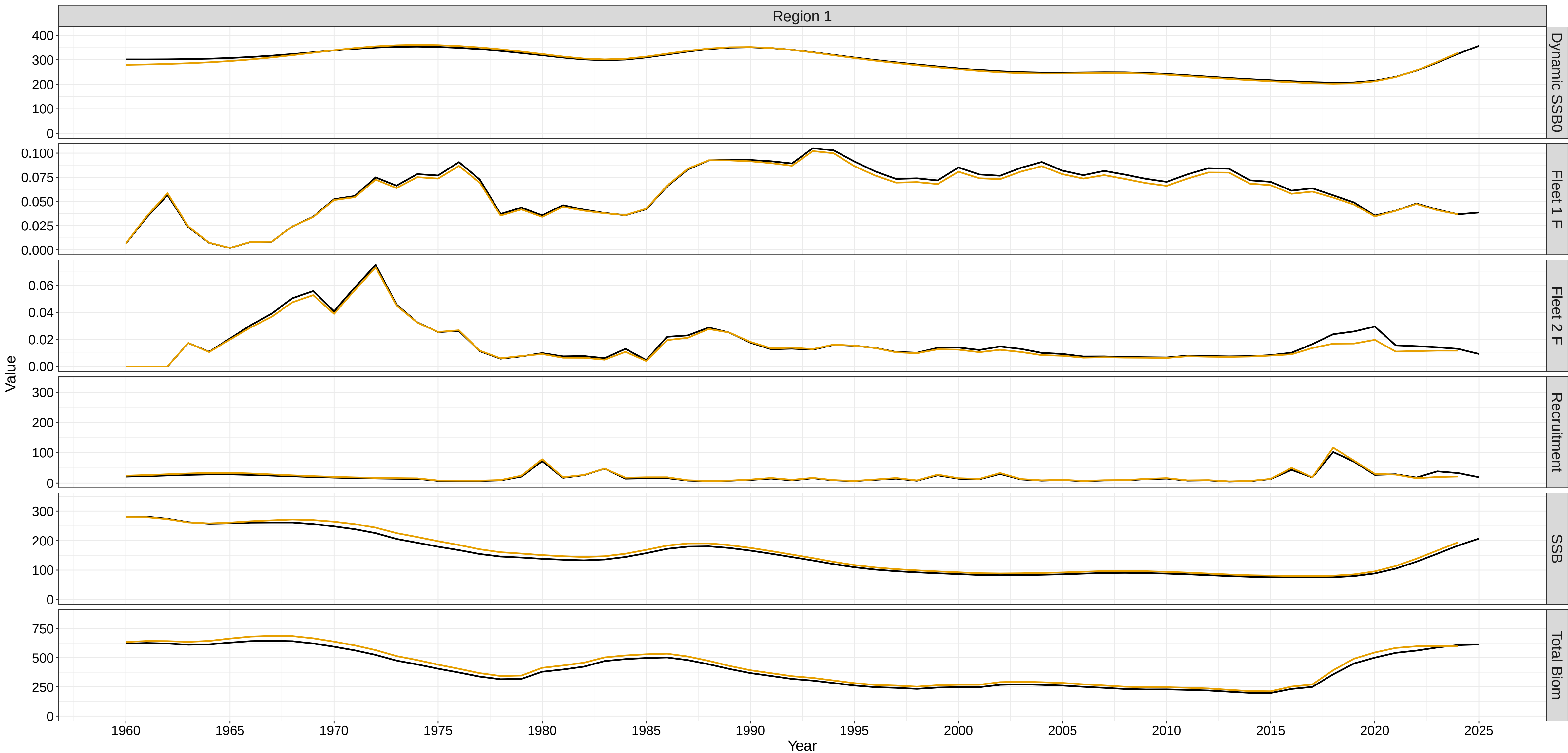
Model — 25.12\_1Reg\_Cont - - - 25\_12\_Drop\_TS\_Upd\_M Year 20 40 60





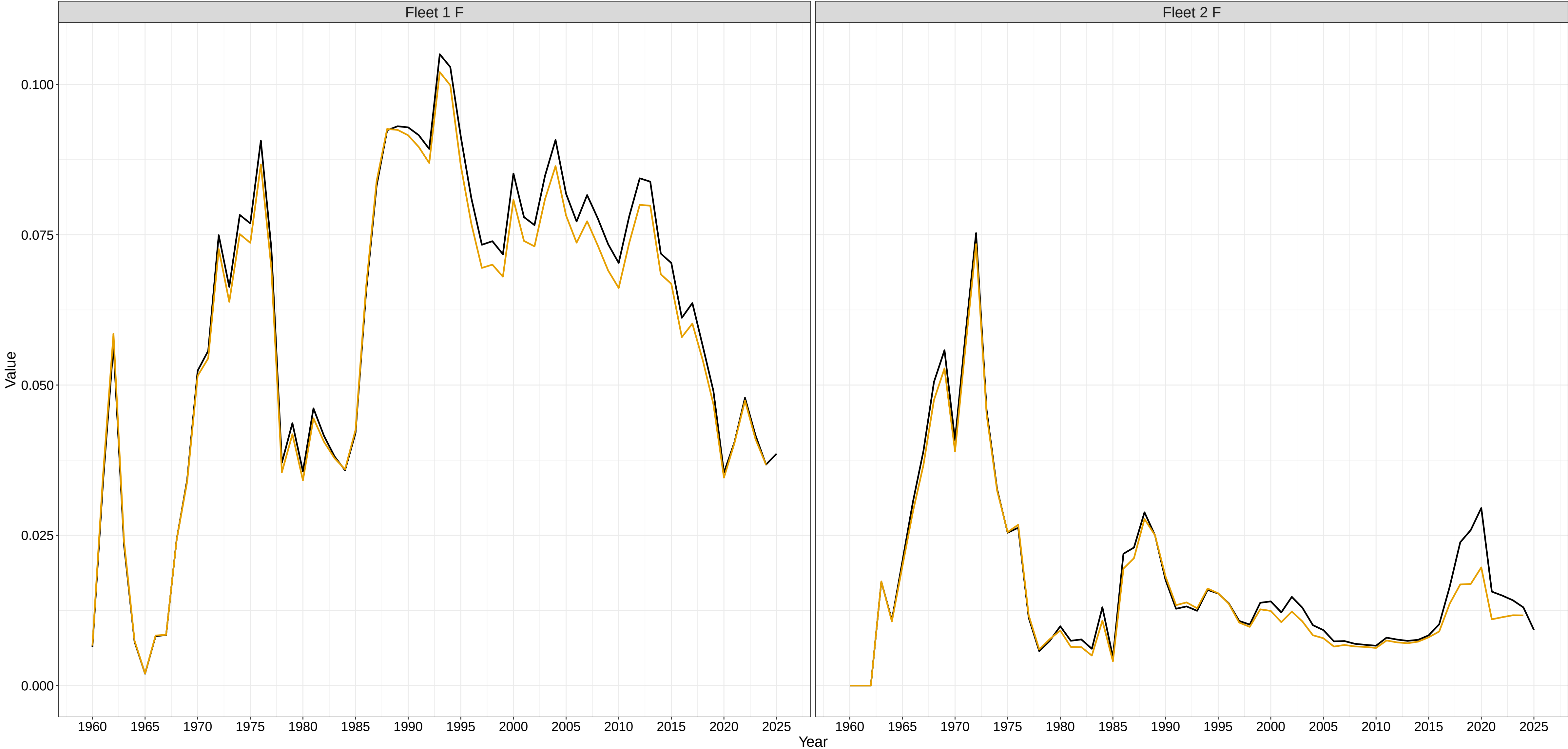
# Time Series Plots

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M

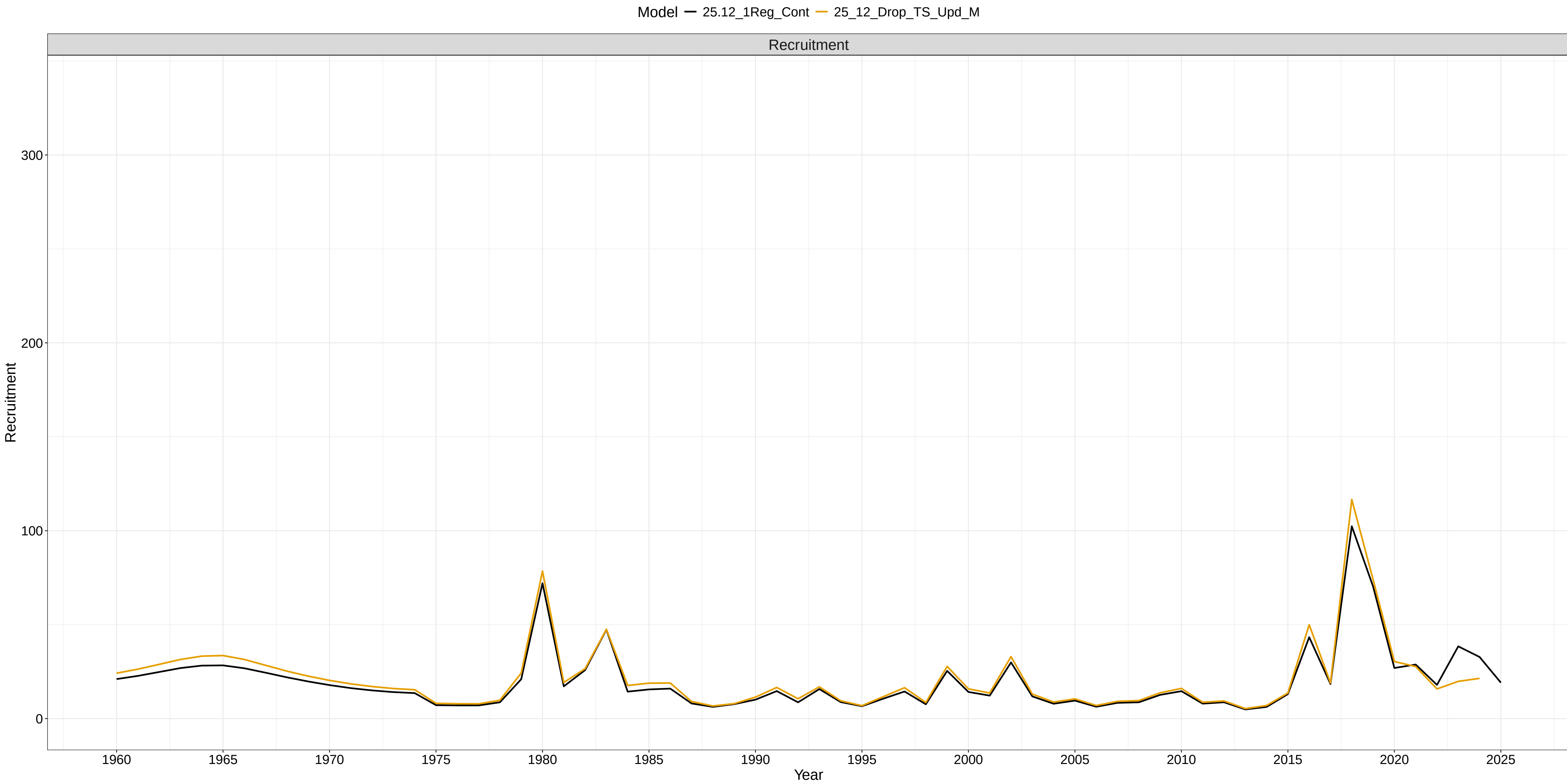


# Fishing Mortality

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M

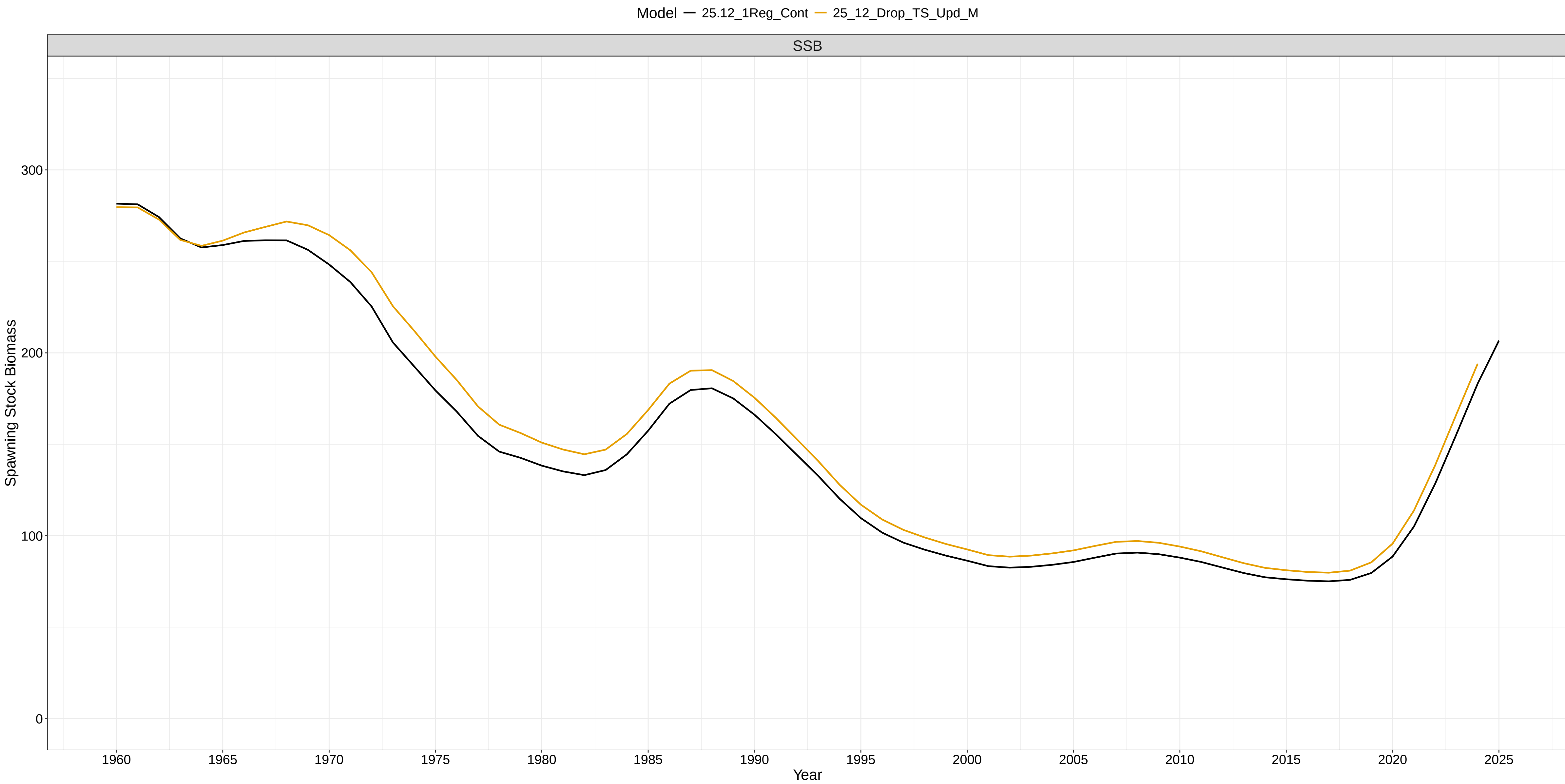


# Recruitment

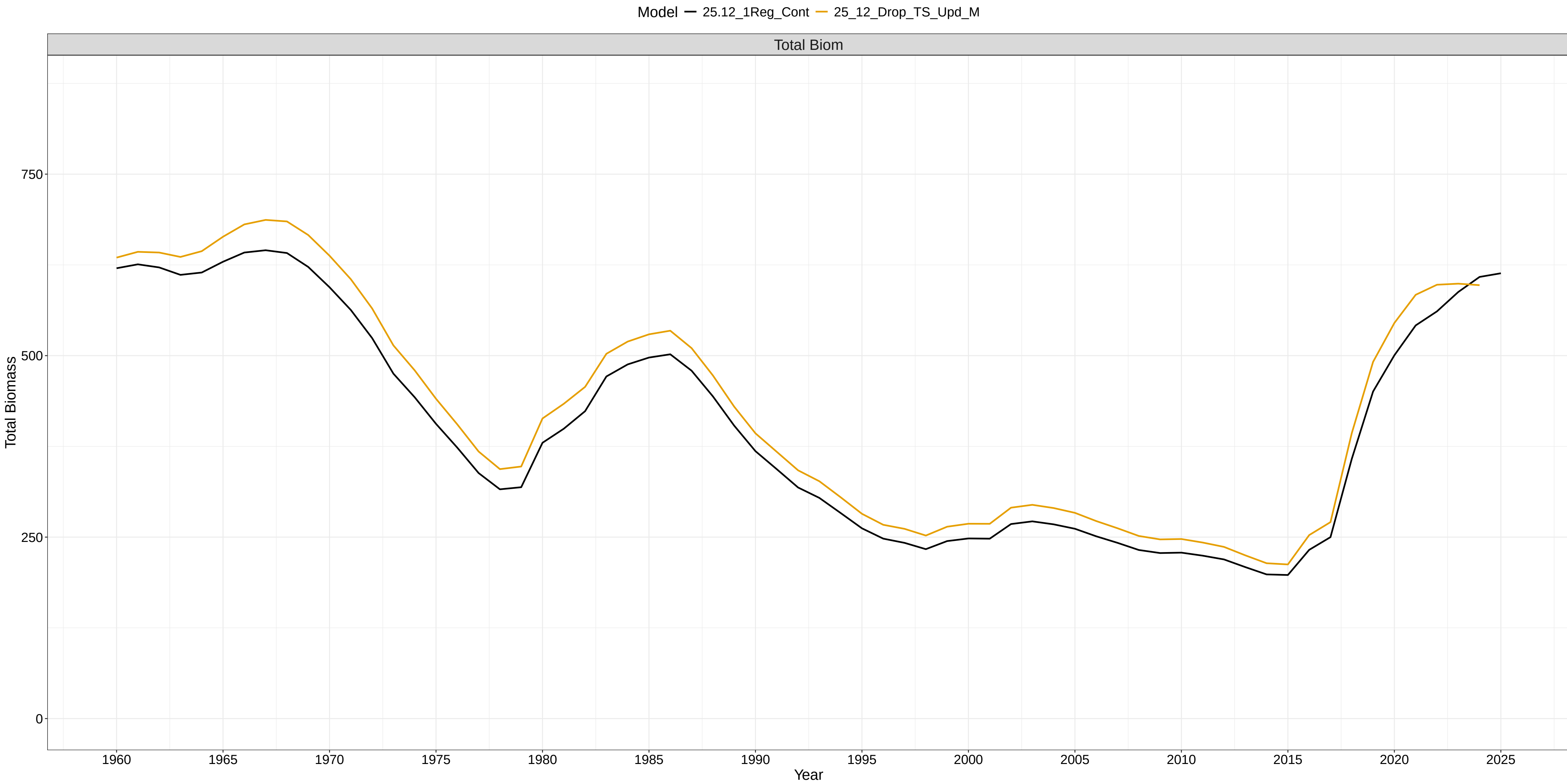




SSB



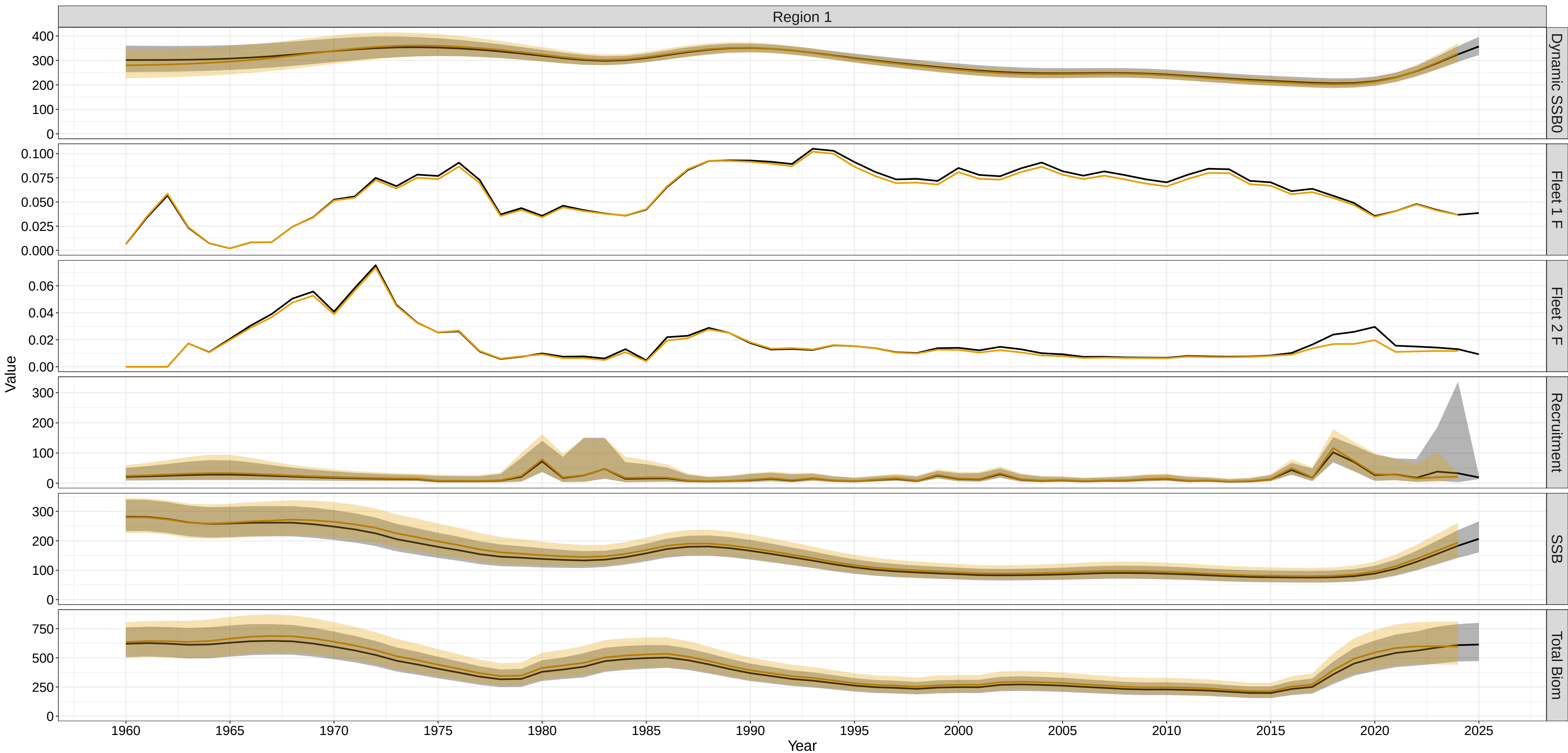
# Biomass





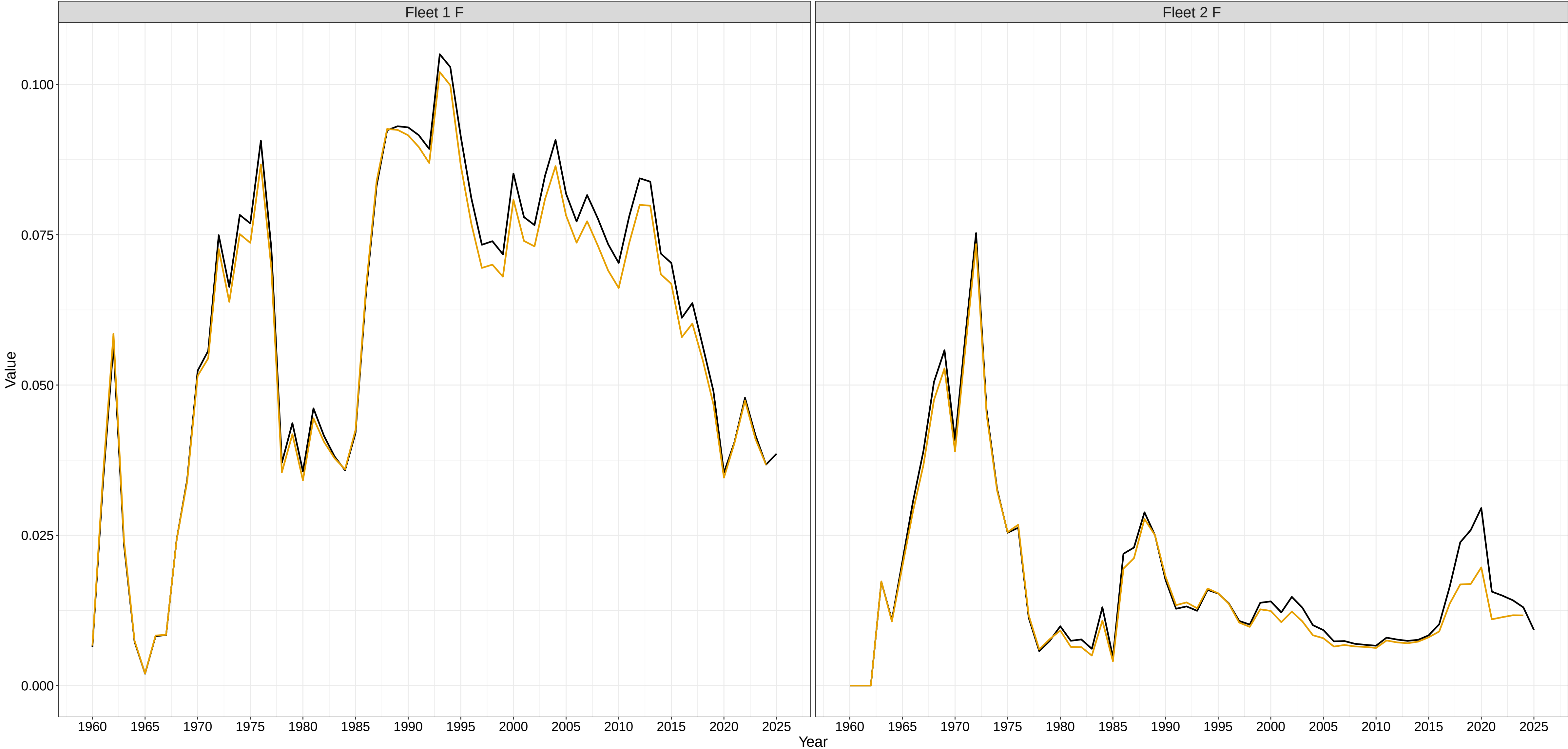
# Time Series Plots

Model — 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M

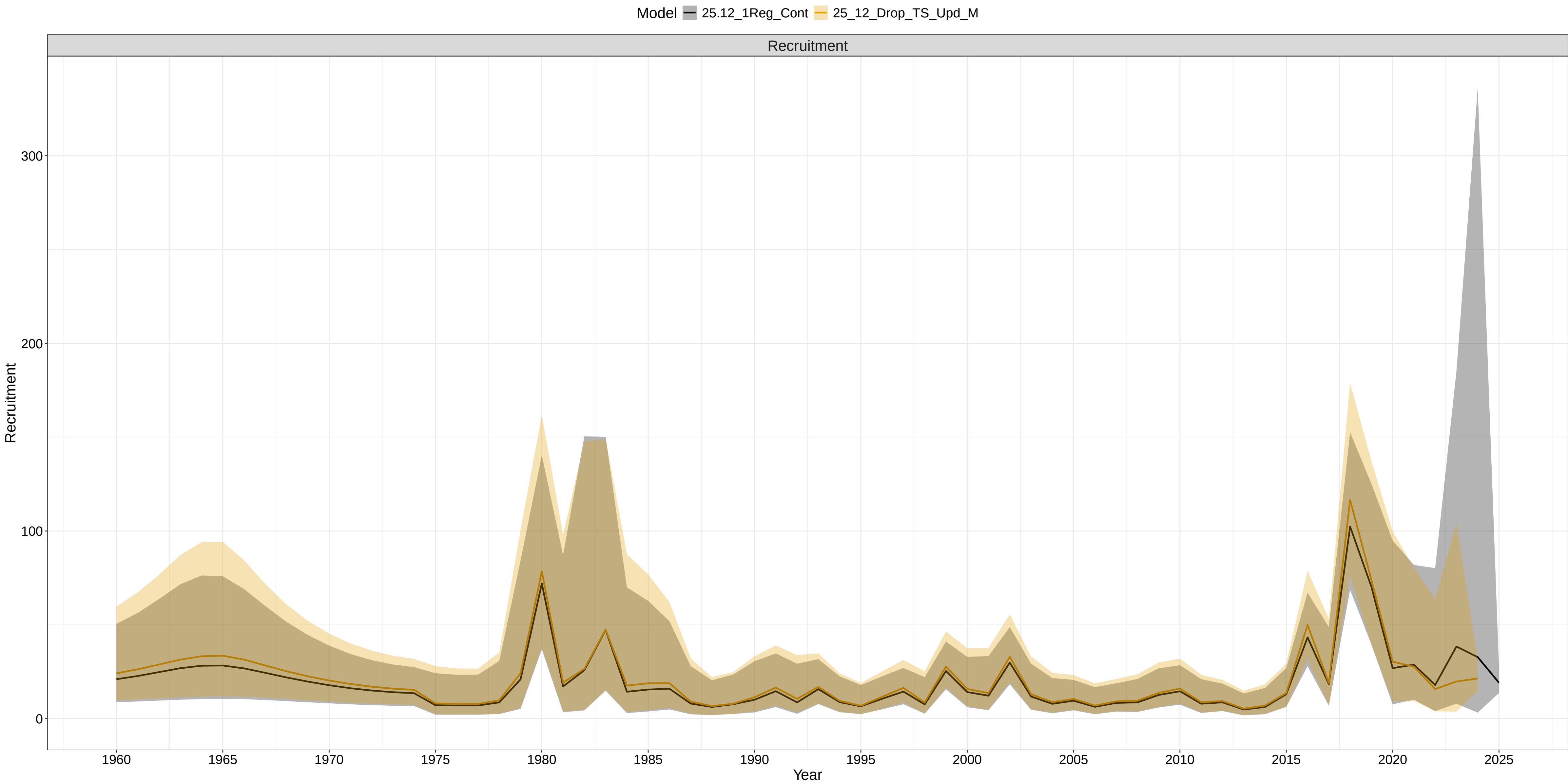


# Fishing Mortality

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M

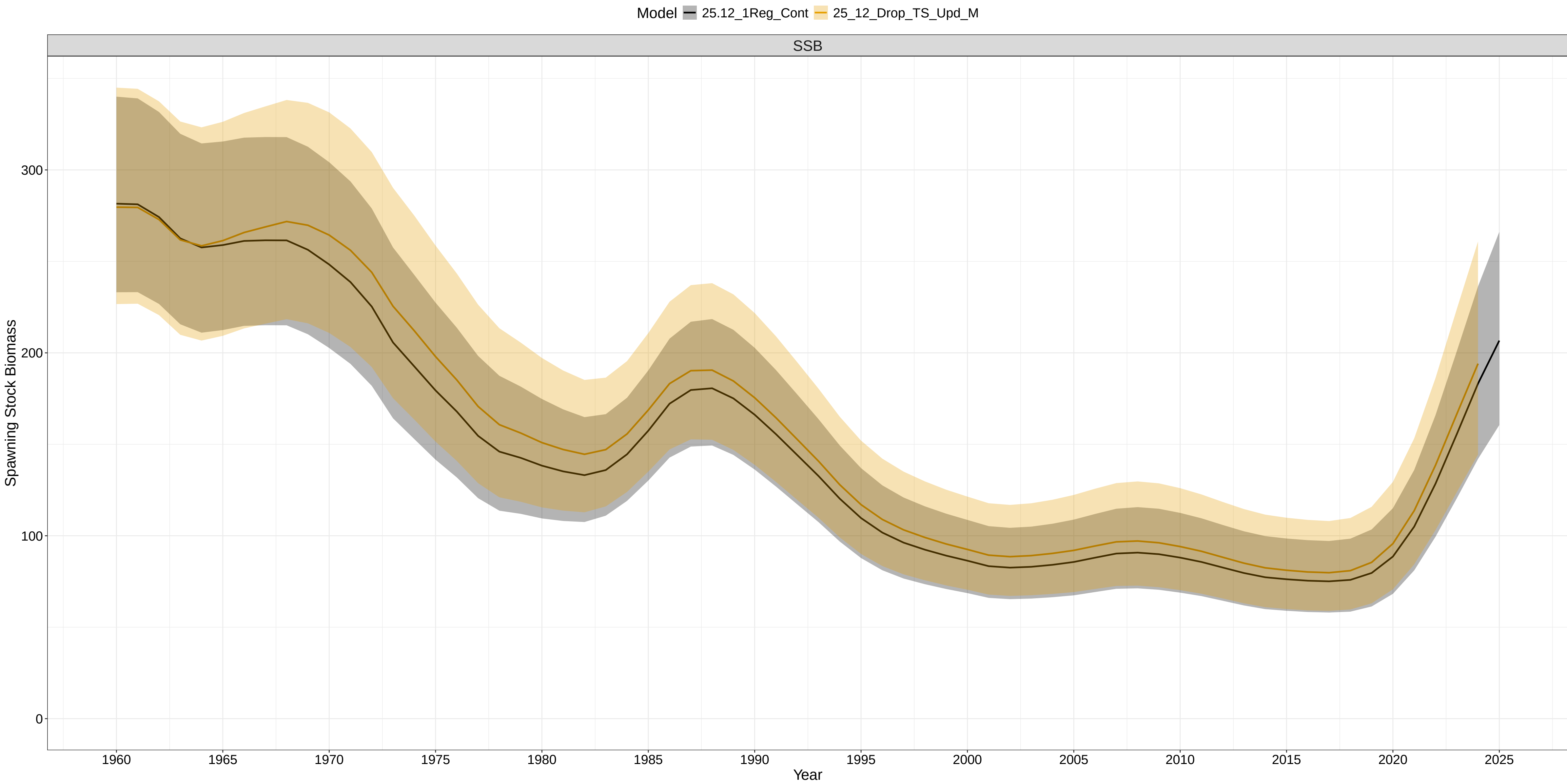


# Recruitment





SSB



# Biomass

